

1° VEME tutorial for the R package **seraphim** 1.0

Plotting the dispersal history of viral lineages

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The present tutorial describes how to use the R package “**seraphim**” [1, 2] to exploit phylogenetically informed movement data obtained through continuous phylogeographic reconstruction [3] in order to plot the dispersal history of Yellow fever viral lineages in Brazil [4]. See also the package manual for further detail on its different functions.

The R package “**seraphim**” is hosted on GitHub (<https://github.com/sdellicour/seraphim>) and the first step is to install it using the “`install_github()`” function of the “**devtools**” package:

```
> install.packages("devtools"); library(devtools)
> install_github("sdellicour/seraphim/unix_OS") # for Unix systems
> install_github("sdellicour/seraphim/windows") # for Windows systems
```

Note that the installation of “**seraphim**” requires the preliminary installation of the following R packages: “**ape**”, “**doMC**” (only available for Unix systems), “**fields**”, “**gdistance**”, “**HDIInterval**”, “**ks**”, “**phytools**”, “**raster**”, “**RColorBrewer**”, “**R.utils**”, and “**vegan**”. Once installed, the package has to be loaded as follows:

```
> library(seraphim)
```

This tutorial requires a Yellow fever virus (YFV) example file also available on the GitHub repository of the package “**seraphim**”: “**YFV_MCC.tree**”, a maximum consensus tree (MCC) tree file retrieved and annotated from 1,000 spatially annotated phylogenetic trees sampled from the post-burn-in posterior distribution of trees inferred for the YFV data set published by Faria and colleagues [4] using the continuous phylogeographic method of Lemey *et al.* [3]. A detailed procedure on how to prepare and conduct a continuous phylogeographic reconstruction using the relaxed random walk (RRW) diffusion model [3] implemented in the software package BEAST 1.10 [5] is available on the BEAST community website (https://beast.community/workshop_continuous_diffusion_yfv).

In the context of this tutorial, we will also need the additional package “**diagram**”. This package can be installed and loaded using the following command lines in R:

```
> install.packages("diagram")
> library(diagram)
```

This tutorial describes how to use the “spreadGraphic1” function to retrieve and display the highest density posterior (HPD) regions associated with internal nodes and reflecting the uncertainty associated with the Bayesian continuous phylogeographic inference. Specifically, these HPD regions will be retrieved from the annotations embedded in a maximum clade consensus (MCC) tree that was obtained and annotated with the program TreeAnnotator (<https://beast.community/treeannotator>). See also, for instance, the programs SPREAD 4 [7] and spread.gl [8] for a very similar approach.

Step 1: extracting spatio-temporal information embedded in the MCC tree

Spatio-temporal information embedded in the MCC tree can be extracted using the “readAnnotatedNexus()” function that was previously available in the package “OutbreakTools” [6]:

```
> mcc_tre = readAnnotatedNexus("YFV_MCC.tree")
```

Before proceeding to the extraction step, we also need to specify the date of the most recent sample as follows:

```
> mostRecentSamplingDatum = 2017.304
```

We can then perform the extraction of the spatiotemporal information embedded in the MCC, which will be saved in a matrix where each row gathers the data related to a distinct phylogeny branch. We will do that with the “mccTreeExtractions()” function applied as follows:

```
> mcc_tab = mccTreeExtraction(mcc_tre, mostRecentSamplingDatum)
> write.csv(mcc_tab, "YFV_MCC.csv", row.names=F, quote=F)
> mcc_tab = read.csv("YFV_MCC.csv", head=T)
```

Step 2: retrieving the HPD region associated with MCC tree internal nodes

We will here extract the 80% HPD region associated with each internal node of the MCC tree retrieved and annotated by the program TreeAnnotator, which can be performed as follows using the “spreadGraphic1()” function:

```
> polygons = suppressWarnings(spreadGraphic1(mcc_tre, mcc_tab))
```

Step 3: defining the different colour scales to use

Here, we will define two distinct sets of colours based on the same colour scale “RdYlGn” generated by the “RColorBrewer” package: (i) “endYears.colours” defining the colour of each node in the MCC tree (i.e. the colour assigned to the descendent node of each branch summarised in “mcc.tab”), and (ii) “polygons.colours” defining the colour of each HPD polygon in the “polygons” list of spatial polygon data frames.

```
> colour_scale = colorRampPalette(brewer.pal(11,"RdYlGn"))(141)[21:121]
> minYear = 2016; maxYear = mostRecentSamplingDatum
> endYears_indices = (((mcc_tab[, "endYear"]-minYear)/(maxYear-minYear))*100)+1
> endYears_colours = colour_scale[endYears_indices]
> polygons_colours = rep(NA, length(polygons))
> for (i in 1:length(polygons)) {
>   date = as.numeric(names(polygons[[i]]))
>   polygon_index = round((((date-minYear)/(maxYear-minYear))*100)+1)
>   polygons_colours[i] = paste0(colour_scale[polygon_index], "40")
> }
```

Step 4: co-plotting the HPD regions and MCC tree

For this final step, we first need to load a template raster that will serve as background to plot the dispersal history of viral lineages, as well as the subnational admin-1 borders of Brazil. These administrative borders were downloaded from the GADM database (<https://gadm.org>) and are here directly cropped using the extent of the template raster:

```
> borders = shapefile("Brazil_admin_1/Brazil_admin_1.shp")
```

HPD regions and MCC tree can then be co-plotted using the script below. This script uses the “curvedarrow()” function from the package “diagram” to plot phylogeny branches as curves (and you can edit its “curve” parameter to play with the strength of curves, “curve” set to “0” returning straight line segments). Note that the script below shows how to plot tree nodes and HPD regions following the same colour scale. The result is displayed in Figure 1.

```
> dev.new(width=5.75, height=6.5)
> par(mar=c(0,0,0,0), oma=c(0,0,0,0), mgp=c(0,0.4,0), lwd=0.4, bty="o")
> plot(borders, col="gray95")
> for (i in 1:length(polygons)) {
>   plot(polygons[[i]], axes=F, col=polygons_colours[i], add=T, border=NA)
> }
> plot(borders, add=T, lwd=0.1, border="gray10")
> for (i in 1:dim(mcc_tab)[1]) {
>   curvedarrow(cbind(mcc_tab[i, "startLon"], mcc_tab[i, "startLat"]),
>               cbind(mcc_tab[i, "endLon"], mcc_tab[i, "endLat"]), arr.length=0,
>               arr.width=0, lwd=0.4, lty=1, lcol="gray10", arr.col=NA,
>               arr.pos=F, curve=0.1, dr=NA, endhead=F)
> }
> for (i in dim(mcc_tab)[1]:1) {
>   if (i == 1) {
```

```

>         points(mcc_tab[i,"startLon"], mcc_tab[i,"startLat"], pch=16,
>               col=colour_scale[1], cex=0.8)
>         points(mcc_tab[i,"startLon"], mcc_tab[i,"startLat"], pch=1,
>               col="gray10", cex=0.8)
>     }
>     points(mcc_tab[i,"endLon"], mcc_tab[i,"endLat"], pch=16,
>           col=endYears_colours[i], cex=0.8)
>     points(mcc_tab[i,"endLon"], mcc_tab[i,"endLat"], pch=1,
>           col="gray10", cex=0.8)
> }
> rast = raster(matrix(nrow=1, ncol=2))
> rast[1] = min(mcc_tab[, "startYear"])
> rast[2] = max(mcc_tab[, "endYear"])
> plot(rast, legend.only=T, add=T, col=colour_scale, legend.width=0.5,
>       legend.shrink=0.3, smallplot=c(0.50,0.90,0.12,0.13), legend.args=list(text="",
>       cex=0.7, line=0.3, col="gray30"), horizontal=T, axis.args=list(cex.axis=0.6,
>       lwd=0, lwd.tick=0.4, tck=-0.8, col.axis="gray30", line=0, mgp=c(0,-0.02,0),
>       at=seq(2016.4,2017.2,0.2)))

```

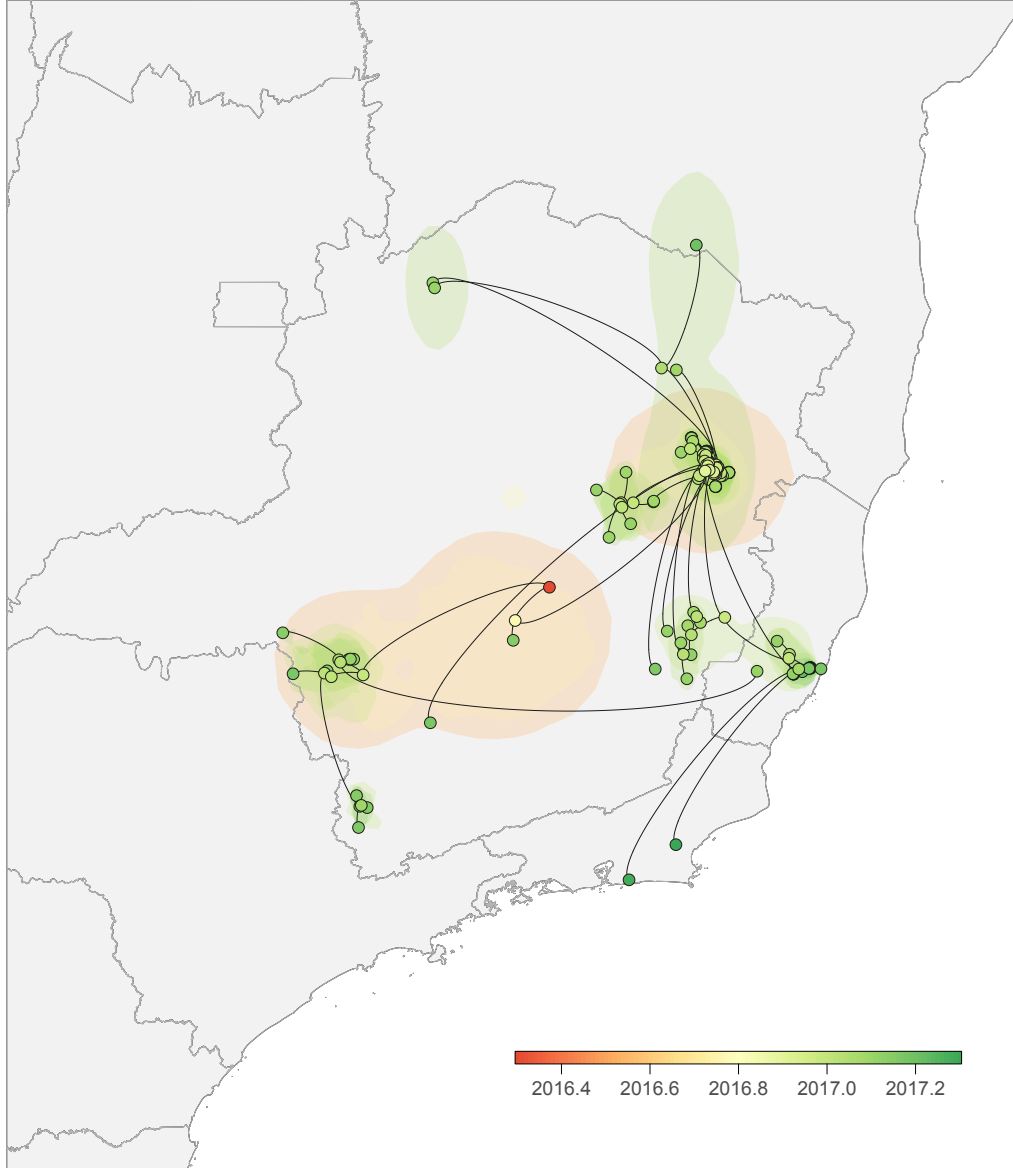


Figure 1: reconstructed spatiotemporal dispersal history of YFV lineages in Brazil based on the data set published by Faria *et al.* [4]: maximum clade credibility (MCC) trees and 80% HPD regions associated with internal nodes. Nodes of the MCC tree are coloured according to their time of occurrence, and 80% HPD regions are superimposed using the same colour scale reflecting time.

References

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- [3] Lemey P, Rambaut A, Welch JJ, Suchard MA (2010). Phylogeography takes a relaxed random walk in continuous space and time. *Molecular Biology & Evolution* 27: 1877-1885.
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- [8] Li Y, Bollen N, Hong SL, Brusselmans M, Gambaro F, Suchard MA, Rambaut A, Lemey P, Dellicour S, Baele G (2024). Spread.gl: visualising pathogen dispersal in a high-performance browser application. *Bioinformatics* 40: btae721.